Exploring the Impact of Rebalancing Frequency on Asset Allocation Strategies

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Abstract

This paper examines how rebalancing frequency influences the long-term performance of four portfolio strategies—60/40, 80/20, Permanent, and All-Weather—using monthly data from 1991 to 2024. We compare return, volatility, and Sharpe ratio outcomes under no rebalancing, annual, quarterly, and monthly regimes. Results suggest that the traditional portfolios (60/40, 80/20, permanent) benefit from periodic rebalancing, whereas All-weather strategies may achieve stronger risk-adjusted performance under passive allocation. Fama-French regressions indicate that the 60/40 strategy consistently delivers significant alpha, whereas the 80/20 portfolio's excess returns are more sensitive to asset specification. Diversified portfolios exhibit lower model fit and capture unexplained returns under passive allocation. Robustness tests using alternative asset proxies confirm the main findings. Overall, rebalancing effectiveness depends on portfolio structure, and static approaches may outperform in multi-asset contexts. Future work should explore threshold-based rebalancing, extended factor models, and real-world frictions to refine these insights.

1. Introduction

1.1. Background and Motivation

Portfolio rebalancing represents a core component of long-term investment strategy, serving to realign portfolio weights with target allocations as market conditions evolve (Buetow Jr et al., 2002). While traditional asset allocation strategies—such as the classic 60/40 equity-bond allocation, have served as benchmarks for decades (Arnott & Lovell, 1993), less attention has been given to how the rebalancing frequency impacts long-term performance. Investors and institutions face trade-offs between discipline and flexibility: frequent rebalancing helps control portfolio risk but may limit upside potential (Tsai, 2001). Against a backdrop of increased interest in long-horizon and passive investment strategies, understanding the performance implications of rebalancing frequency has become increasingly relevant.

1.2. Research Objective

This paper examine how different rebalancing frequencies affect portfolio performance across a range of commonly used asset allocation strategies. Using monthly data from 1991 to 2024, we compare annualized returns, volatility, and Sharpe ratios under four rebalancing schemes: no rebalancing, annual, quarterly, and monthly. In addition, we assess whether the Fama-French three-factor model explains the return patterns observed under each strategy-frequency configuration.

1.3. Contribution and Relevance

This research offers several novel insights into the ongoing discourse on optimal portfolio rebalancing practices. First, it offers a comprehensive comparison of multiple rebalancing frequencies across diverse portfolio strategies, including the 60/40, 80/20, Permanent Portfolio, and All Weather frameworks. Second, it links

performance outcomes to factor-based return explanations, offering insight into whether alpha persists after accounting for standard risk exposures. Finally, by incorporating robustness checks using alternative asset inputs, the findings highlight the sensitivity of performance attribution to both asset selection and portfolio management discipline. Findings from this study bear practical relevance for long-horizon investors and may also inform future empirical investigations on rebalancing efficiency.

2. Data and Methodology

2.1. Asset Selection and Description

This study investigates asset allocation strategies constructed from four key asset classes: equities, government bonds, gold, and broad commodities. Equity exposure is proxied by the S&P 500 Total Return Index (SPXT), while fixed income is represented by the Bloomberg Intermediate-Term U.S. Treasury Index (LUATTRUU), which reflects bonds with maturities between 6 to 8 years. Gold exposure is captured using the Bloomberg Commodity Gold Subindex Total Return (BCOMGCTR), and the broader commodity basket is proxied by the S&P GSCI Total Return Index (SPGSCITR). These total return indices incorporate reinvested income or roll yield and thus offer a more realistic representation of investable returns.

Monthly observations from January 1991 to December 2024 are retrieved from Bloomberg. The 3-month U.S. Treasury bill yield (USGG3M Index) serves as a proxy for the risk-free rate in regression analysis but is not deducted again in Sharpe ratio computations to avoid double-counting. For risk-factor regressions, monthly returns

of the Fama-French three factors are sourced from the Kenneth R. French Data Library. Table 1 summarizes the key variables.

Table 1. Description of the key variables

Variable Name	Symbol	Description	Type	Frequency	Source
SP500	SPXT Index	S&P 500 Total Return Index	Equity	Monthly	Bloomberg
Treasury	LUATTRUU Index	Intermediate-Term U.S. Treasury Index	Fixed Income	Monthly	Bloomberg
Gold	BCOMGCTR Index	Bloomberg Commodity Gold Subindex Total Return	Commodity	Monthly	Bloomberg
Commodity	SPGSCITR Index	S&P GSCI Commodity Index Total Return (ex-Gold)	Commodity	Monthly	Bloomberg
RF	USGG3M Index	3-Month U.S. T-Bill Yield	Risk-Free Rate	Monthly	Bloomberg

2.2. Portfolio Construction

Four asset allocation strategies are considered in this study, each representing a distinct risk-return philosophy:

- 60/40 Portfolio: A conventional balanced portfolio allocating 60% to equities and 40% to government bonds.
- 80/20 Portfolio: A more aggressive configuration with 80% equities and 20% bonds.
- Permanent Portfolio: An equally weighted strategy distributing 25% each to equities, bonds, gold, and commodities.
- All-Weather portfolio: A dynamic strategy inspired by the macro diversification philosophy of Bridgewater's All Weather Portfolio (Dalio, 2016), but implemented using the Equal Risk Contribution (ERC) method. Weights are computed monthly using a 12-month rolling covariance matrix of asset returns,

ensuring each asset contributes equally to total portfolio volatility (Maillard et al., 2010). This replaces the prior fixed-weight approximation and allows the strategy to better reflect changing market conditions.

Initial weights are applied at the beginning of the sample period. Portfolios are either held without rebalancing (passive drift) or rebalanced to their original targets on a fixed schedule, depending on the regime.

2.3. Rebalancing Mechanism

This study evaluates four rebalancing regimes: no rebalancing, annual, quarterly, and monthly. Under the no-rebalancing condition, portfolio weights are allowed to drift over time as asset prices fluctuate. This approach reflects a passive, buy-and-hold way, where allocation changes arise solely from market movements.

In contrast, periodic rebalancing resets portfolio weights to their original targets at fixed intervals—either annually, quarterly, or monthly. This time-based, deterministic strategy is commonly adopted in empirical finance for its simplicity and ease of implementation (Bányai et al., 2024). At each rebalancing point, the portfolio is adjusted using updated market values observed at the end of the period. Unlike threshold-based or opportunistic frameworks, which rely on deviation triggers, the present study focuses exclusively on time-based rebalancing to isolate the effect of frequency. Following common practice in the literature, transaction costs, taxes, and other frictions are excluded to ensure a clean comparison of portfolio dynamics (Daryanani, 2008).

2.4. Evaluation Metrics

Portfolio performance is evaluated using both absolute and risk-adjusted return measures. The primary metric is annualized return, computed by compounding monthly returns over the sample period. To assess portfolio risk, we calculate the annualized volatility, defined as the standard deviation of monthly returns scaled by the square root of 12.

To capture risk-adjusted performance, we compute the Sharpe ratio as the difference between the portfolio's annualized return and the annualized risk-free rate, divided by the portfolio's annualized volatility. Since all portfolio returns are based on total return indices, which already include dividends, coupon income, and roll yield, the risk-free rate is not subtracted again to avoid double-counting. For consistency, Sharpe ratios are computed across all strategies and rebalancing configurations using monthly return data annualized where applicable. The 3-month U.S. Treasury bill yield (USGG3M Index) is still retained as a proxy for the risk-free rate in Fama-French regressions and other benchmark comparisons.

In addition to these summary statistics, we conduct Fama-French three-factor regressions to evaluate whether observed returns can be attributed to standard risk exposures. The model estimates alpha (intercept), market beta, size beta, and value beta for each portfolio, alongside the R-squared to assess the overall fit. This methodology helps distinguish return components driven by systematic risk factors from those representing unexplained excess returns (Fama & French, 2023).

3. Empirical Analysis

3.1. Performance of Buy-and-Hold Strategies

To establish a benchmark for evaluating the impact of rebalancing, this section presents the performance of four commonly referenced portfolio strategies under a buy-and-hold framework. The strategies examined include the classic 60/40 and 80/20 stock-bond allocations, as well as the Permanent Portfolio and the All Weather Portfolio. Each strategy was initialized in January 1991 with its target weightings and allowed to drift naturally without any rebalancing throughout the sample period ending in December 2024.

Figure 1. Cumulative Net Value of Buy-and-Hold Portfolios (1991–2024)

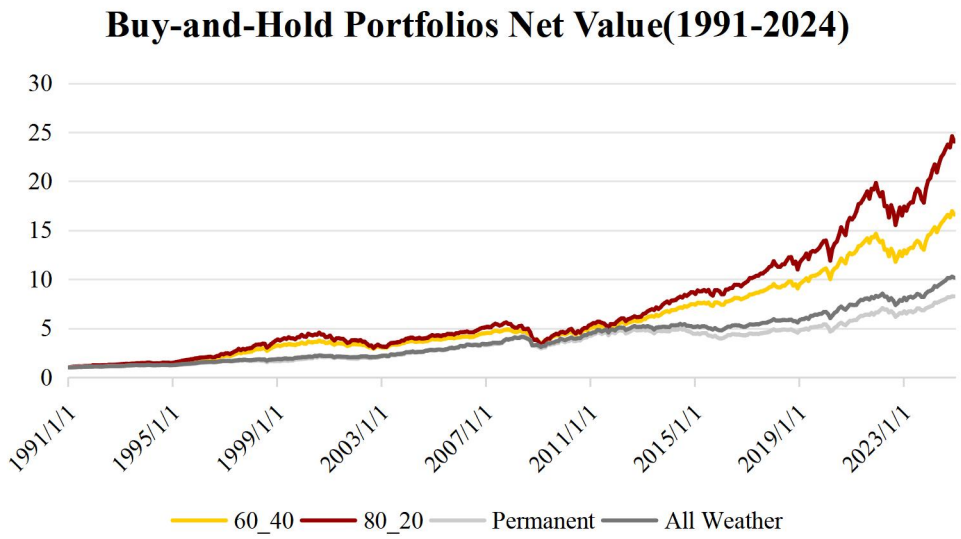


Figure 1 illustrates the cumulative net value growth of each portfolio, beginning from an initial value of 1. The 80/20 portfolio achieved the highest terminal value, followed closely by the 60/40 portfolio, reflecting the long-term strength of equity performance over the 35-year period. Meanwhile, the All Weather and Permanent portfolios delivered more stable growth paths with lower volatility, consistent with their diversified and risk-balanced structures.

Table 2. Performance Metrics of Buy-and-Hold Strategies (1991–2024)

Strategy	Annualized Return (%)	Annualized Volatility (%)	Sharpe Ratio
60/40	12.76	10.26	**0.999**
80/20	**14.90**	**13.58**	0.913
Permanent	**9.61**	10.29	**0.691**
All-Weather	10.50	**9.03**	0.885

Table 2 summarizes key performance metrics for the buy-and-hold portfolios. While the 80/20 portfolio achieved the highest average return (14.90%), it also exhibited the greatest volatility. In contrast, the All Weather portfolio delivered a more moderate return (10.50%) but achieved the lowest volatility (9.03%) among all strategies. Its Sharpe ratio (0.885), indicating a favorable balance between return and risk. These results serve as a performance benchmark against which the effectiveness of rebalancing strategies will be assessed in the subsequent sections.

3.2. Performance of Rebalanced Portfolios

Building on the results of the buy-and-hold analysis, this section assesses the performance of the same four portfolio strategies under periodic rebalancing. Each portfolio is rebalanced back to its target allocation at fixed frequencies—annually, quarterly, or monthly—throughout the sample period from January 1991 to December 2024. The rebalancing mechanism reflects a realistic implementation approach: portfolio weights are allowed to drift between rebalancing dates, with adjustments made only at predetermined intervals based on updated asset prices.

Figure 2. Annualized Return by Strategy and Rebalancing Frequency (1991–2024)

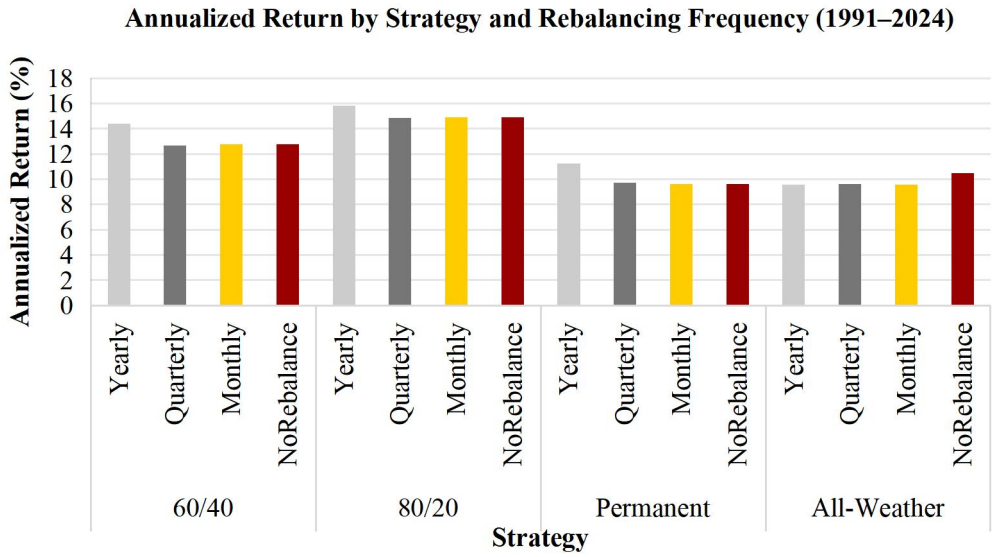


Figure 2 and Table 3 jointly show the impact of rebalancing frequency on portfolio performance across four strategies from 1991 to 2024. The results reveal that the Yearly rebalanced 80/20 portfolio achieved the highest annualized return 15.28%), albeit with elevated volatility. For the 60/40 portfolio, Quarterly rebalancing offered the best risk-adjusted performance, with the highest Sharpe ratio (1.001).

Interestingly, the All-Weather portfolio without rebalancing outperformed all other configurations within its group, achieving both the highest return (10.50%) and Sharpe ratio (0.885), suggesting that over-diversified portfolios may benefit from a hands-off approach. Conversely, for equity-dominated strategies, modest periodic rebalancing, particularly yearly, helps balance return and risk effectively.

These findings imply that rebalancing is not a one-size-fits-all solution. While it helps contain risk and maintain strategic asset allocations, its effect on performance is nuanced and strategy-dependent. Investors should tailor their rebalancing frequency based on their portfolio's equity exposure and return objectives.

Table 3. Performance Summary of Rebalanced Portfolios (1991–2024)

Strategy	Frequency	Annualized Return (%)	Annualized Volatility (%)	Sharpe Ratio
60/40	Yearly	**14.42**	**12.88**	**0.925**
	Quarterly	**12.67**	**10.16**	**1.001**
	Monthly	12.76	10.26	0.999
	NoRebalance	12.76	10.26	0.999
80/20	Yearly	**15.85**	**15.12**	**0.883**
	Quarterly	**14.84**	**13.50**	**0.913**
	Monthly	14.90	13.58	0.913
	NoRebalance	14.90	13.58	0.913
Permanent	Yearly	**11.24**	**10.84**	**0.806**
	Quarterly	9.71	**10.20**	0.706
	Monthly	**9.61**	10.29	**0.691**
	NoRebalance	9.61	10.29	0.691
All-Weather	Yearly	**9.57**	**10.20**	**0.693**
	Quarterly	9.61	9.93	0.716
	Monthly	9.58	9.93	0.713
	NoRebalance	**10.50**	**9.03**	**0.885**

3.3. Factor-Based Performance Evaluation

To evaluate whether the observed portfolio performance stems from exposure to systematic risk factors rather than genuine outperformance, this section employs a three-factor regression analysis based on the original Fama-French (1993) model. Specifically, we regress the monthly excess returns of each portfolio against the market risk premium (MKT-RF), the size factor (SMB), and the value factor (HML), using factor data obtained from the updated Fama-French data library:

$$R_t^p - R_{f,t} = \alpha + \beta_1(R_{m,t} - R_{f,t}) + \beta_2 SMB_t + \beta_3 HML_T + \varepsilon_t$$

where R_t^p denotes the portfolio return, $R_{f,t}$ the risk-free rate, and α captures the portfolio's abnormal return not explained by standard risk factors.

Monthly factor data are sourced from the Fama-French U.S. Research Data Library, covering the full sample period from February 1991 to December 2024. Excess

returns are calculated by subtracting the one-month Treasury bill rate from the raw portfolio's monthly return. The regression is conducted separately for each rebalanced strategy.

Table 4. Fama-French 3-Factor Regression Results (1991–2024)

Strategy	Frequency	Alpha(%)	Beta_MKT	Beta_SMB	Beta_HML	R_squared
60/40	Yearly	**0.09****	0.5744	-0.1259	-0.0056	0.9431
	Quarterly	0.08**	0.582	-0.1287	-0.0067	0.9485
	Monthly	0.08**	0.5841	-0.1329	-0.0077	0.9536
	NoRebalance	**0.07****	0.5857	-0.1314	-0.005	0.954
80/20	Yearly	**0.05****	0.7793	-0.1499	0.0112	0.9858
	Quarterly	0.04*	0.7851	-0.152	0.0103	0.987
	Monthly	0.04*	0.7869	-0.1549	0.0096	0.9883
	NoRebalance	**0.04**	0.7881	-0.1539	0.0115	0.9884
Permanent	Yearly	**0.02**	0.3438	0.021	0.071	0.3598
	Quarterly	0.01	0.3488	0.0249	0.0718	0.3812
	Monthly	0.01	0.3509	0.0209	0.0747	0.3848
	NoRebalance	**0**	0.3537	0.0207	0.0786	0.3876
All-Weather	Yearly	**0.02**	0.3563	0.02	0.0765	0.39
	Quarterly	0.04	0.349	0.0103	0.0662	0.3899
	Monthly	0.04	0.3471	0.087	0.0648	0.3853
	NoRebalance	**0.06**	0.3804	-0.0167	0.0359	0.5273

Note: *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 4 illustrates that the 60/40 portfolio consistently demonstrates statistically significant positive alpha across all rebalancing frequencies, with the yearly-rebalanced version achieving the highest alpha of 0.09% per month (significant at the 5% level). The 80/20 portfolio also exhibits positive alpha, though with slightly lower magnitudes and significance.

In contrast, the Permanent and All-Weather portfolios generally show no significant alpha under most configurations, indicating that their returns are well explained by exposure to systematic risk factors. However, the All-Weather strategy without rebalancing stands out, generating a positive and statistically significant alpha

(0.06%), suggesting that static allocation in diversified portfolios may allow for harvesting additional sources of return beyond the traditional Fama-French factors.

Regarding factor loadings, all portfolios show positive market beta, with higher sensitivity observed in equity-heavy portfolios such as 80/20. The negative SMB loadings in 60/40 and 80/20 suggest a tilt toward large-cap stocks, while the HML coefficients are close to zero across strategies, implying limited exposure to the value-growth dimension.

The explanatory power of the model is notably high for the 60/40 and 80/20 strategies ($R^2 > 0.94$), whereas the Permanent and All-Weather portfolios have lower R^2 values (as low as 0.36), indicating that their performance may be influenced by other non-modeled factors, such as momentum or macroeconomic risk exposures..

3.4. Robustness Test

To assess the robustness of the findings, we replicate the strategy construction using alternative asset proxies. Specifically, we replace the Treasury component with a longer-duration bond index and substitute the original commodity series with the BCOM index. The revised portfolios are constructed following the same methodology, covering the period from January 1991 to December 2024, and evaluated under four rebalancing schemes: annual, quarterly, monthly, and no rebalancing.

Table 5. Performance Summary of Rebalanced Portfolios (1991–2024) in the Robustness

Strategy	Frequency	Annualized Return (%)	Annualized Volatility (%)	Sharpe Ratio
60/40	Yearly	**9.22**	**9.42**	**0.712**
	Quarterly	9.18	9.48	0.703
	Monthly	**9.10**	9.50	0.694
	NoRebalance	9.12	**9.52**	**0.693**
80/20	Yearly	**10.76**	**12.16**	**0.678**
	Quarterly	10.74	12.22	0.673
	Monthly	**10.69**	12.37	0.668
	NoRebalance	10.69	**12.52**	**0.667**
Permanent	Yearly	**6.80**	**8.40**	**0.512**
	Quarterly	6.71	8.35	0.502
	Monthly	6.64	**8.34**	0.494
	NoRebalance	**6.63**	8.35	**0.493**
All-Weather	Yearly	**6.71**	**8.37**	**0.502**
	Quarterly	6.75	8.25	0.513
	Monthly	6.82	8.26	0.522
	NoRebalance	**6.97**	**8.18**	**0.545**

In Table 5, the 60/40 and 80/20 portfolios continued to show strong performance, with the yearly-rebalanced 80/20 strategy delivering the highest return (10.76%) and the 60/40 portfolio achieving its best Sharpe ratio (0.712) under yearly rebalancing. The All-Weather strategy, once again, exhibited its best performance without rebalancing (Sharpe ratio of 0.545), reaffirming that a static allocation approach may be more effective for highly diversified portfolios.

These results confirm that the main conclusions are robust to changes in underlying asset proxies. While rebalancing enhances risk-adjusted performance in moderate-risk portfolios like 60/40, equity-heavy or highly diversified strategies exhibit distinct and consistent responses to rebalancing, independent of specific asset choices.

Table 6. Fama-French 3-Factor Regression Results (1991–2024) in the Robustness Test

Strategy	Frequency	Alpha(%)	Beta_MKT	Beta_SMB	Beta_HML	R_squared
60/40	Yearly	**0.06***	0.6063	-0.1172	0.0093	0.9573
	Quarterly	0.06*	0.6116	-0.1192	0.009	0.9598
	Monthly	0.05	0.6137	-0.1231	0.0077	0.9632
	NoRebalance	**0.05**	0.6151	-0.122	0.0098	0.9634
80/20	Yearly	**0.03**	0.796	-0.1458	0.0187	0.9879
	Quarterly	0.03	0.8	-0.1472	0.0183	0.9885
	Monthly	0.02	0.8017	-0.15	0.0175	0.9894
	NoRebalance	**0.02**	0.8028	-0.1493	0.0189	0.9895
Permanent	Yearly	**0.06**	0.347	-0.0095	0.052	0.4099
	Quarterly	0.05	0.3515	-0.0072	0.0538	0.4246
	Monthly	0.04	0.3525	-0.0102	0.0541	0.4271
	NoRebalance	**0.04**	0.354	-0.0098	0.0561	0.4301
All-Weather	Yearly	**0.04**	0.3569	-0.0067	0.0611	0.437
	Quarterly	0.05	0.3492	-0.0091	0.0535	0.429
	Monthly	**0.06**	0.3502	-0.0100	0.0545	0.430
	NoRebalance	0.04	0.3869	-0.0236	0.0489	0.531

Note: *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 6 presents the Fama-French 3-factor regression results under the robustness test. The 60/40 portfolio retained statistically significant positive alpha under both yearly and quarterly rebalancing (0.06%), reaffirming the robustness of its outperformance after controlling for standard risk factors.

In contrast, the alpha of the 80/20 portfolio lost statistical significance across all rebalancing schemes, suggesting that its previously observed excess return was more sensitive to the choice of underlying asset proxies. This indicates a weaker robustness relative to the 60/40 strategy.

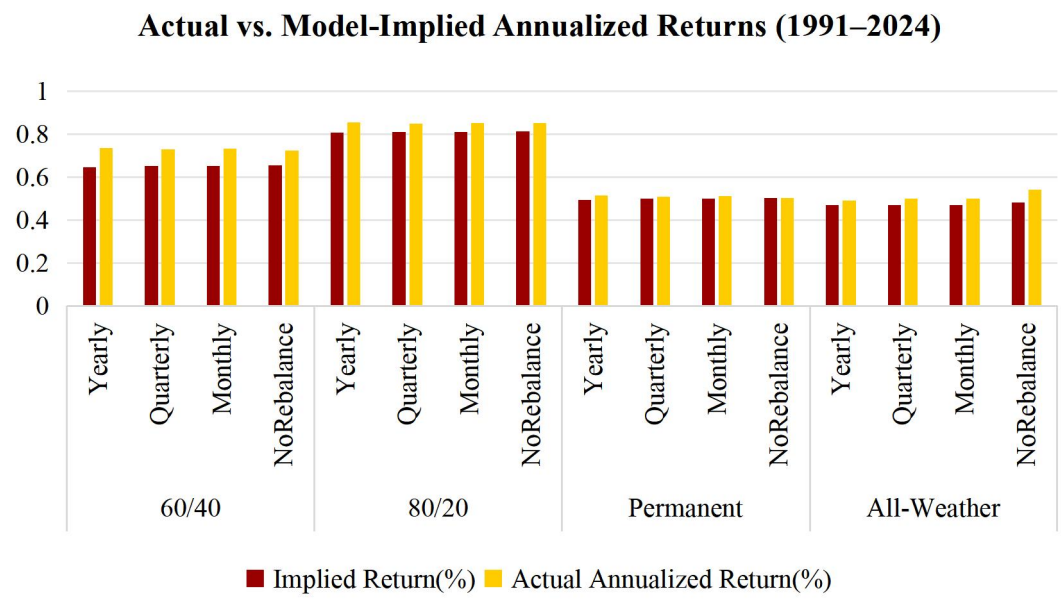
Interestingly, the Permanent portfolio exhibited slightly higher alpha in the robustness test, with the yearly-rebalanced version achieving 0.06% (significant at the 10% level). The All-Weather portfolio maintained stable but insignificant alpha values, consistent with its design as a risk-balanced and factor-diversified allocation.

Overall, these results suggest that the 60/40 strategy's performance remains robust to variations in asset selection, while the 80/20 portfolio's excess returns may be more contingent on specific asset choices. The Permanent and All-Weather strategies, though less reliant on traditional equity factors, continue to show stable behavior under asset substitutions.

3.5. Implied Expected Returns from the Fama-French Model

To complement the regression analysis, this section examines the extent to which the Fama-French three-factor model explains the observed returns across all portfolio configurations. Specifically, we compute model-implied annualized returns by combining each portfolio's estimated alpha and factor loadings with the historical average monthly returns of the market, size, and value factors over the 1991–2024 period (Figure 3).

Figure 3. Actual vs. Model-Implied Annualized Returns (1991–2024).



For the 60/40 and 80/20 strategies, the Fama-French model demonstrates a strong

ability to replicate actual performance. The close alignment between implied and observed returns—particularly for the 80/20 portfolio—suggests that their performance is largely driven by exposure to systematic market, size, and value factors, leaving limited unexplained alpha.

In contrast, the Permanent and All-Weather portfolios consistently exhibit higher realized returns than those implied by the three-factor model. This discrepancy is especially pronounced under no-rebalancing schemes, indicating that these strategies may capture additional sources of return beyond the scope of traditional equity risk factors. Such findings suggest the presence of alternative return drivers—such as cross-asset risk premiums, inflation sensitivity, or momentum—that are not accounted for in the Fama-French specification.

Overall, the results reinforce that while the Fama-French model is effective for traditional equity-heavy portfolios, it underexplains returns in more diversified or multi-asset strategies, thereby motivating the exploration of extended factor models in future research.

4. Conclusion

This study evaluates how rebalancing frequency affects portfolio performance across four asset allocation strategies over the 1991–2024 period. Results show that the benefits of rebalancing are strategy-dependent. For tradition portfolios like 60/40 and 80/20, periodic rebalancing, particularly annual or quarterly, enhances returns and improves risk-adjusted outcomes. In contrast, All-Weather model with risk-dominant weighting perform best with minimal intervention, with no-rebalancing configurations

often delivering the highest Sharpe ratios.

Fama-French regressions confirm these patterns. The 60/40 strategy consistently generates significant alpha, while the 80/20 portfolio's alpha is more sensitive to asset specification. The All-Weather strategy shows meaningful unexplained returns only in its passive form, suggesting exposure to alternative risk sources. Robustness checks with alternative asset proxies uphold the core findings.

Model-implied returns closely approximate realized outcomes for equity-centric portfolios, but consistently underpredict those of diversified strategies, underscoring the limitations of standard factor models.

Future research could extend this work by incorporating higher-frequency data, additional pricing factors, threshold-based rebalancing rules, and real-world constraints such as transaction costs or regime shifts.

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Appendix

A1. Descriptive Statistics of Asset Prices

Asset	Mean	Std Dev	Min	25th Percentile	Median	75th Percentile	Max
SPXT Index	3092.48	2866.55	335.95	1298.2	1938.48	3892.34	13227.13
LUATTRUU Index	1537.12	641.78	457.23	958.24	1490.10	2151.55	2605.74
BCOMGCTR Index	232.52	148.77	69.04	95.32	183.50	335.55	720.66
SPGSCITR Index	3636.52	1527.39	1349.82	2408.78	3280.88	4509.94	10560.00
LBUSTRUU Index	1378.50	612.62	390.43	825.51	1316.71	1974.29	2396.78
BCOM Index	113.43	30.58	60.90	90.67	103.78	130.49	233.03
USGG3M Index	2.72	2.29	-0.02	0.18	2.36	4.99	8.04

A2. Portfolio Configurations

Strategy	Equities	Bonds	Gold	Commodities
60/40	60%	40%	-	-
80/20	80%	20%	-	-
Permanent	25%	25%	25%	25%
All-Weather	ERC	ERC	ERC	ERC

A3. Descriptive Statistics of Fama-French Three-Factor Data

Factor	mean	std	min	max
MKT_RF	0.74	4.42	-17.23	13.65
SMB	0.07	3.17	-17.2	21.36
HML	0.13	3.28	-13.88	12.8
RF	0.22	0.19	0	0.69